

Inflationary Observables from Dimensional Flow in Scale-Dependent Geometric Field Theory

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Abstract. In Scale-Dependent Geometric Field Theory (SDGFT), inflation is not driven by an ad-hoc scalar field but by the dimensional flow from the UV fixed point $D_{UV} = 2$ to the IR attractor $D^* = 67/24$. The resulting $f(R) = R^{67/48}$ action uniquely determines all slow-roll parameters.

We derive the primordial spectral index $n_s = 1 - 2(2n - 1)/[N_e(2n - 1) + n] \approx 0.967$ and the tensor-to-scalar ratio $r \approx 0.013$ from the Mukhanov–Sasaki equation in the Jordan frame, using the exact e-fold count $N_e \approx 59.95$ predicted by the lattice topology.

These results are consistent with Planck 2018 CMB data ($n_s = 0.9649 \pm 0.0042$, within 0.5σ) and lie well below the BICEP/Keck upper limit ($r < 0.036$).

Erratum: We explicitly retract a formula $n_s = 1 + n_f^R/67 = 1.0126$ appearing in earlier preprints, which predicted an unphysical blue spectrum excluded at $> 11\sigma$ by Planck data. The error originated from a coordinate projection inconsistency in the mapping from spectral to effective dimension. The correct R^n slow-roll analysis presented here yields a red tilt in excellent agreement with observation.

Keywords: inflation, spectral index, tensor-to-scalar ratio, $f(R)$ gravity, modified gravity, 24-cell, SDGFT

1 Introduction

The inflationary paradigm [1, 2, 3] provides a compelling explanation for the homogeneity, isotropy, and spatial flatness of the observable universe, as well as the origin of primordial density perturbations. In the standard treatment, inflation is driven by a slowly rolling scalar field (the inflaton) with a suitably flat potential. While phenomenologically successful, this framework carries the burden of an ad-hoc inflaton sector: neither the identity of the scalar field nor the shape of its potential is derived from first principles.

Scale-Dependent Geometric Field Theory (SDGFT) [4, 5] offers an alternative mechanism: inflation arises from the *dimensional flow* of spacetime itself. In the UV, the spectral dimension approaches $D_{UV} = 2$; in the IR, it relaxes to the unique attractor $D^* \approx 2.797$, proven to be a Banach contraction fixed point [5]. This dimensional transition generates the required quasi-exponential expansion without an explicit inflaton.

The gravitational dynamics during the dimensional flow are governed by the modified action $f(R) = R^n$ with $n = D^*/2 = 67/48$ [7]. This belongs to the well-studied class of R^n inflation theories [10, 11, 12], but with the crucial distinction that in SDGFT the exponent n is *not a free parameter*—it is rigidly fixed by the 24-cell topology.

In this paper we derive the complete set of inflationary observables (§2–4), present a formal erratum correcting a legacy formula (§5), and confront all predictions with current CMB data (§??).

2 E-Folds from the Dimensional Gap

The number of inflationary e-folds is determined by the “dimensional gap” between the UV starting point and the IR attractor, normalized by the two fundamental lattice scales:

$$N_e = \frac{D^*}{\Delta} \ln \left[\frac{D^* - 2 - \delta}{\Delta \delta} \right]. \quad (1)$$

The UV starting dimension $D_{\text{UV}} = 2$ appears implicitly through the numerator $D^* - 2 - \delta$ of the logarithmic argument: this is the residual dimensional gap after subtracting the UV fixed point and the lattice tension correction.

Substituting the canonical values $D^* = D_{\text{fp}}^* \approx 2.79676$, $\Delta = 5/24$, and $\delta = 1/24$:

$$N_e \approx \frac{2.79676}{0.20833} \ln \left[\frac{2.79676 - 2 - 0.04167}{0.20833 \times 0.04167} \right] = 13.424 \times \ln(86.90) \approx 59.95. \quad (2)$$

This falls squarely within the standard range $50 \lesssim N_e \lesssim 60$ required to solve the horizon and flatness problems, without any tuning.

3 Slow-Roll Parameters

3.1 General R^n slow-roll analysis

For a general $f(R) = R^n$ action in the Jordan frame, the slow-roll parameters were derived by Hwang and Noh [11]:

$$\epsilon = \frac{(2n-1)(n-2)^2}{[N_e(2n-1) + n]^2}, \quad (3)$$

$$\eta = \frac{(2n-1)(2n^2 - 7n + 4)}{[N_e(2n-1) + n]^2} - \frac{2n-1}{N_e(2n-1) + n}. \quad (4)$$

3.2 SDGFT predictions

With $n = 67/48$ and $N_e \approx 59.95$, the denominator evaluates to

$$N_e(2n-1) + n = 59.95 \times \frac{43}{24} + \frac{67}{48} \approx 107.67 + 1.396 = 109.06. \quad (5)$$

The slow-roll parameters are then:

$$\epsilon = \frac{\frac{43}{24} \times \left(\frac{67}{48} - 2\right)^2}{109.06^2} = \frac{1.7917 \times 0.03057}{11893.8} \approx 5.52 \times 10^{-5}, \quad (6)$$

$$\eta \approx -0.0168. \quad (7)$$

Both $|\epsilon| \ll 1$ and $|\eta| \ll 1$, confirming the validity of the slow-roll approximation.

4 Primordial Observables

4.1 Spectral index

The primordial scalar spectral index is related to the slow-roll parameters by

$$n_s = 1 - 6\epsilon + 2\eta. \quad (8)$$

For R^n gravity, this simplifies to the compact form

$$n_s = 1 - \frac{2(2n-1)}{N_e(2n-1) + n}. \quad (9)$$

Numerical evaluation yields

$$n_s = 1 - \frac{2 \times 1.7917}{109.06} = 1 - 0.03287 = 0.96713. \quad (10)$$

4.2 Tensor-to-scalar ratio

The tensor-to-scalar ratio in R^n gravity is

$$r = \frac{48(2n-1)^2}{[N_e(2n-1) + n]^2}. \quad (11)$$

Note that the standard single-field consistency relation $r = 16\epsilon$ is *not* satisfied in $f(R)$ gravity. Instead, one has

$$r = \frac{48(2n-1)}{(n-2)^2} \epsilon, \quad (12)$$

so the ratio $r/\epsilon \approx 236$ is much larger than 16.

Numerically:

$$r = \frac{48 \times 3.2102}{11893.8} \approx 0.01295. \quad (13)$$

4.3 Summary of predictions

Table 1: SDGFT inflationary predictions compared to Planck 2018 and BICEP/Keck observations. All predictions follow from $n = 67/48$ and $N_e \approx 59.95$ with no free parameters.

Observable	SDGFT	Observed	Agreement
N_e	59.95	50–60	consistent
n_s	0.9671	0.9649 ± 0.0042	0.5σ
r	0.0130	< 0.036 (95% CL)	consistent
ϵ	5.52×10^{-5}	< 0.002	consistent
η	-0.0168	$ \eta < 0.1$	consistent
β_{iso}	$1/36 \approx 0.028$	< 0.038 (95% CL)	consistent

The isocurvature amplitude $\beta_{\text{iso}} = 1/36$ arises from the six-cone geometry: the observer samples one of six independent channels, yielding an amplitude fraction $(1/6)^2 = 1/36$ in power [8].

5 Erratum: Retraction of the Blue Spectrum Formula

5.1 The erroneous formula

In earlier preprints (Paper 04/24 and related drafts) [6], a formula for the spectral index was stated as

$$n_s^{(\text{legacy})} = 1 + \frac{n_f^R}{67} = 1 + \frac{0.845}{67} \approx 1.0126. \quad (14)$$

This predicted a *blue* spectrum ($n_s > 1$), implying that primordial perturbations grow toward smaller scales—a prediction dramatically at odds with all CMB observations.

5.2 Quantitative falsification

The Planck 2018 constraint [13] is $n_s = 0.9649 \pm 0.0042$ (68% CL, TT,TE,EE+lowE+lensing). The blue prediction deviates by

$$\frac{|n_s^{(\text{legacy})} - n_s^{(\text{Planck})}|}{\sigma_{n_s}} = \frac{|1.0126 - 0.9649|}{0.0042} = \frac{0.0477}{0.0042} \approx 11.4\sigma. \quad (15)$$

This constitutes an overwhelming rejection. For comparison, the correct red prediction $n_s = 0.9671$ deviates by only 0.5σ .

5.3 Origin of the error

The erroneous formula arose from a coordinate projection inconsistency in the mapping between the spectral dimension flow and the effective gravitational dimension. Specifically, the parameter n_f^R was computed in a co-moving frame where the D -dependent corrections enter with the *wrong sign*, because the Jacobian of the frame transformation was neglected.

The correct derivation proceeds entirely within the Jordan frame using the Hwang–Noh slow-roll parameters (Eqs. 3–4), yielding the red tilt $n_s \approx 0.967$ presented in §4.

5.4 Implications

This correction has no impact on the other sectors of SDGFT (particle physics, cosmological densities, gauge couplings), since these do not depend on the inflationary slow-roll parameters. The tree-level dimension $D_{\text{tree}}^* = 67/24$, the $f(R)$ exponent $n = 67/48$, and the Banach fixed-point proof [5] remain unchanged.

6 Comparison with Starobinsky Inflation

The most natural comparison is with Starobinsky R^2 inflation [3], which corresponds to $n = 2$ in our notation. Table 2 compares the two.

Table 2: Comparison of SDGFT $R^{67/48}$ inflation with Starobinsky R^2 inflation ($N_e = 60$).

	SDGFT ($n = 67/48$)	Starobinsky ($n = 2$)
n_s	0.9671	0.9667
r	0.0130	0.0033
r/ϵ	≈ 236	≈ 12
α_M	$19/86 \approx 0.221$	$1/3$
α_T	0	0
Free parameters	0	1 (M_{scalaron})

While the spectral indices are nearly identical, the tensor-to-scalar ratios differ by a factor of 4. This is because the SDGFT exponent $n = 67/48 \approx 1.396$ is closer to 1 (GR) than $n = 2$, reducing the effective scalaron mass and enhancing the relative tensor amplitude. The prediction $r \approx 0.013$ is within the target sensitivity of both LiteBIRD [15] ($\sigma_r \approx 0.001$) and CMB-S4 [16] ($\sigma_r \approx 0.003$), making this a *decisive discriminator* between SDGFT and Starobinsky inflation.

7 Falsifiability and Future Tests

The zero-parameter nature of SDGFT inflation generates sharp, falsifiable predictions:

1. **Tensor-to-scalar ratio:** A measurement of r inconsistent with 0.013 ± 0.003 would falsify the theory. LiteBIRD (launch ~ 2032) and CMB-S4 (~ 2029) will reach the required sensitivity.
2. **Spectral running:** The running $dn_s/d \ln k$ is predicted to be $\approx -(2n-1)/[N_e(2n-1)+n]^2 \approx -1.5 \times 10^{-4}$, consistent with zero running and indistinguishable from Starobinsky at current precision.
3. **Isocurvature:** The isocurvature fraction $\beta_{\text{iso}} = 1/36 \approx 0.028$ is below but close to the current Planck 95% CL upper limit of 0.038. Improvement by a factor of 2 in sensitivity could detect this signal.

8 Conclusions

We have derived the complete set of inflationary observables in SDGFT from the unique gravitational action $f(R) = R^{67/48}$, with the exponent rigidly fixed by the 24-cell lattice topology. The key results are:

- $N_e \approx 59.95$ e-folds, predicted by the dimensional gap formula without tuning.
- $n_s \approx 0.967$, consistent with Planck 2018 at 0.5σ .
- $r \approx 0.013$, consistent with BICEP/Keck and within reach of LiteBIRD and CMB-S4.
- A formal retraction of the legacy blue spectrum formula $n_s = 1.0126$, excluded at $> 11\sigma$.

The theory makes a crisp, falsifiable prediction: the tensor-to-scalar ratio is $r = 0.013$, a factor of 4 larger than Starobinsky's R^2 inflation. Forthcoming CMB polarization experiments will discriminate between these scenarios within the next decade.

Acknowledgments

All numerical results in this paper have been verified against the open-source SDGFT Python framework [9], which includes dedicated unit tests for inflationary observables (spectral index, tensor-to-scalar ratio, slow-roll parameters, and blue spectrum falsification) as part of a 1115-test validation suite.

References

- [1] A. H. Guth, Phys. Rev. D **23**, 347 (1981).
- [2] A. D. Linde, Phys. Lett. B **108**, 389 (1982).
- [3] A. A. Starobinsky, Phys. Lett. B **91**, 99 (1980).

- [4] D. Goldstein, “Axiomatic Foundations of SDGFT,” SDGFT-2026-01 (in preparation).
- [5] D. Goldstein, “Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT,” SDGFT-2026-04 (companion paper).
- [6] D. Goldstein, “Effective Spectral Dimension in SDGFT,” preprint v1 (2024, superseded).
- [7] D. Goldstein, “The $f(R) = R^{67/48}$ Action and Horndeski Parameters,” SDGFT-2026-14 (in preparation).
- [8] D. Goldstein, “Neutrino Isocurvature and the S_8 Tension,” SDGFT-2026-22 (in preparation).
- [9] D. Goldstein, `sdgft` Python package, <https://github.com/TODO/sdgft> (2026).
- [10] V. Müller, H.-J. Schmidt, and A. A. Starobinsky, *Class. Quant. Grav.* **7**, 1163 (1990).
- [11] J. Hwang and H. Noh, *Phys. Lett. B* **506**, 13 (2001).
- [12] L. Amendola, R. Gannouji, D. Polarski, and S. Tsujikawa, *Phys. Rev. D* **75**, 083504 (2007).
- [13] Planck Collaboration (Y. Akrami *et al.*), *Astron. Astrophys.* **641**, A10 (2020).
- [14] BICEP/Keck Collaboration (P. A. R. Ade *et al.*), *Phys. Rev. Lett.* **127**, 151301 (2021).
- [15] LiteBIRD Collaboration (E. Allys *et al.*), *Prog. Theor. Exp. Phys.* **2023**, 042F01.
- [16] CMB-S4 Collaboration (K. N. Abazajian *et al.*), *Astrophys. J.* **926**, 54 (2022).